SVGAP11-CXSV Clean Package Manual

Yi Liao @ scau.edu.cn

2026-07-07

Contents

1 SVGAP11-CXSV Clean Package Manual **2**
1.1 1. What was fixed and cleaned . . . 2
1.1.1 1.1 Parallel::ForkManager warning fixed . . . 2
1.1.2 1.2 Author line added . . . 2
1.1.3 1.3 Non-production files moved to trash . . . 2
1.2 2. Clean package structure . . . 3
1.3 3. Input preparation: ISO1 reference + A3/A4/A5 queries . . . 3
1.3.1 3.1 Recommended directory layout . . . 3
1.3.2 3.2 g.list . . . 3
1.3.3 3.3 Genome FASTA names . . . 4
1.3.4 3.4 AXT file names . . . 4
1.4 4. Running the full pipeline . . . 4
1.4.1 4.1 Recommended mode: start from existing AXT . . . 4
1.4.2 4.2 Start from MAF . . . 5
1.4.3 4.3 Start from aligner outputs . . . 5
1.4.4 4.4 Use true reciprocal alignments . . . 5
1.5 5. Main output directories . . . 5
1.6 6. Detailed workflow steps and parameters . . . 6
1.6.1 Step 01: 01_Convert2Axt.pl . . . 6
1.6.2 Step 02: 02_ChainNetSyn.pl . . . 7
1.6.3 Step 03: 03_SynNetFilter.pl . . . 7
1.6.4 Step 04: 04_PairwiseSV.pl . . . 8
1.6.5 Step 05: 05_CombineSV.pl . . . 9
1.6.6 Step 06: 06_genotype_DEL.pl . . . 10
1.6.7 Step 07: 07_genotype_INS.pl . . . 10
1.6.8 Step 08: 08_genotype_small_indels.pl . . . 11
1.6.9 Step 09: 09_SNPgenotyping.pl . . . 12
1.6.10 Step 10: 10_SVannotation.pl . . . 12
1.6.11 Step 11: 11_INVgenotyping.pl . . . 13
1.6.12 Step 12: 12_combine_genotype_CXSV.pl . . . 14
1.6.13 SVGAP_pipeline.pl controller options . . . 14
1.6.14 SVGAP_vcf_tools.pl . . . 15
1.7 7. Output sanity checks . . . 16
1.8 8. Troubleshooting . . . 16
1.8.1 8.1 Parallel::ForkManager warning still appears . . . 16
1.8.2 8.2 AXT files are not detected . . . 16
1.8.3 8.3 Genome FASTA not found . . . 17
1.8.4 8.4 minimap2 is missing . . . 17
1.8.5 8.5 Memory and threads . . . 17
1.8.6 8.6 qnet versus reciprocal alignments . . . 17
1.9 9. Recommended training workflow . . . 17

1 SVGAP11-CXSV Clean Package Manual

Author tag: every Perl script in this package now contains `Author: Yi Liao @ scau.edu.cn`.
Purpose: run assembly-to-assembly SV discovery, merging, complex-SV handling, SNP/small-InDel genotyping and optional annotation from one reference assembly and multiple query assemblies.
Example dataset: `IS01` is the reference genome; `A3`, `A4` and `A5` are query genomes.

1.1 1. What was fixed and cleaned

1.1.1 1.1 Parallel::ForkManager warning fixed

The recurring warning was:

```
WARNING: Parallel::ForkManager is not installed; falling back to single-thread mode
```

The package already includes the bundled module:

```
lib/Parallel/ForkManager.pm
```

and its dependencies. The problem was that some scripts did not consistently add the package-local `lib/` directory to Perl `@INC`. Therefore, `require Parallel::ForkManager` failed and the scripts fell back to single-thread mode.

All `.pl` scripts now include package-local library paths through `FindBin`, for example:

```
use FindBin qw($Bin);
use lib "$Bin/lib";
use lib "$Bin/pub";
use lib "$Bin/../lib";
use lib "$Bin/../pub";
use lib "$Bin/../../lib";
use lib "$Bin/../../pub";
```

As a result, the package-local `Parallel::ForkManager` is used automatically. Users normally do not need to install this Perl module manually.

Quick test:

```
cd SVGAP11_clean_fixed
perl -Ilib -MParallel::ForkManager -e 'print "Parallel::ForkManager OK\n"'
```

1.1.2 1.2 Author line added

Every Perl script now contains:

```
# Author: Yi Liao @ scau.edu.cn
```

1.1.3 1.3 Non-production files moved to trash

The following files were moved to `trash/`, not deleted:

```
trash/benchmark_tools/
  simulate_population_variants_benchmark_shared.v4.pl
  13_compare_truth_SVGAP.pl
  14_diagnose_small_indel_benchmark.pl

trash/integration_notes/
  SVGAP_pipeline.add_12_CXSV.patch
  README_CXSV_integration.txt

trash/legacy_utilities/
  FaFileChecl.pl
```

```
  INS2TDP.pl
  Mafreformat.pl
  SplitFa.pl
```

They are not required for running the production ISO1/A3/A4/A5 workflow. They can be restored if simulation, benchmarking or legacy utility behavior is needed.

1.2 2. Clean package structure

```
SVGAP11_clean_fixed/
  SVGAP_pipeline.pl                 # Main controller for the full workflow
  01_Convert2Axt.pl                 # Step 01: convert alignments to AXT
  02_ChainNetSyn.pl                 # Step 02: AXT -> chain/net/synnet
  03_SynNetFilter.pl                # Step 03: synnet filtering + single-coverage MAF
  04_PairwiseSV.pl                  # Step 04: pairwise SV calling
  05_CombineSV.pl                   # Step 05: merge SV candidates
  06_genotype_DEL.pl                # Step 06: deletion genotyping
  07_genotype_INS.pl                # Step 07: insertion genotyping
  08_genotype_small_indels.pl       # Step 08: small-InDel genotyping
  09_SNPgenotyping.pl               # Step 09: SNP genotyping
  10_SVannotation.pl                # Step 10: SV annotation
  11_INVgenotyping.pl               # Step 11: inversion genotyping
  12_combine_genotype_CXSV.pl       # Step 12: complex-SV merging and genotyping
  SVGAP_vcf_tools.pl                # VCF merging and sample-order utility
  lib/                              # Bundled Perl dependencies, including Parallel::ForkManage
  pub/                              # UCSC/TBA/MUMMER/Stretcher helpers
  util/Test_dependency.pl           # Dependency test script
  docs/                             # Manuals
  trash/                            # Archived non-production files
```

1.3 3. Input preparation: ISO1 reference + A3/A4/A5 queries

1.3.1 3.1 Recommended directory layout

```
project/
  SVGAP11_clean_fixed/
  genomes/
    ISO1.fa
    A3.fa
    A4.fa
    A5.fa
  g.list
  01_axt/Target/                # If starting from AXT
    ISO1vsA3.axt
    ISO1vsA4.axt
    ISO1vsA5.axt
```

1.3.2 3.2 g.list

`g.list` contains one genome ID per line. The reference ID must be included. The pipeline removes the reference from the VCF sample columns, so the final sample order will be `A3, A4, A5`.

```
ISO1
A3
A4
A5
```

1.3.3 3.3 Genome FASTA names

For each genome ID, the scripts search for one of these names:

```
ID
ID.fa
ID.fasta
ID.fna
```

For this example, use:

```
genomes/ISO1.fa
genomes/A3.fa
genomes/A4.fa
genomes/A5.fa
```

Recommended FASTA headers:

```
>ISO1.Chr01
>A3.Chr01
>A4.Chr01
>A5.Chr01
```

Avoid sequence IDs beginning with numbers.

1.3.4 3.4 AXT file names

If starting directly from AXT, forward AXT files must be named:

```
ISO1vsA3.axt
ISO1vsA4.axt
ISO1vsA5.axt
```

If true independently generated reciprocal alignments are available, prepare:

```
A3vsISO1.axt
A4vsISO1.axt
A5vsISO1.axt
```

Do not generate reciprocal AXT by simply reversing forward AXT. That only reorders coordinates; it is not an independent query-vs-reference alignment.

1.4 4. Running the full pipeline

1.4.1 4.1 Recommended mode: start from existing AXT

```
cd project/SVGAP11_clean_fixed
```

```
perl SVGAP_pipeline.pl \
  --refname ISO1 \
  --glist ../g.list \
  --genome_dir ../genomes \
  --start_from axt \
  --axt_dir ../01_axt/Target \
  --outdir ../SVGAP_run \
  --threads 12 \
  --speciesname ISO1_A3_A4_A5 \
  --skip_annotation
```

`--skip_annotation` is recommended for the first training run because annotation may require TE/CDS inputs and a working minimap2 installation. The core SV/SNP/InDel pipeline can be validated first.

1.4.2 4.2 Start from MAF

Forward MAF files should be named like:

```
ISO1vsA3.maf
ISO1vsA4.maf
ISO1vsA5.maf
```

Command:

```
perl SVGAP_pipeline.pl \
  --refname ISO1 \
  --glist ../g.list \
  --genome_dir ../genomes \
  --start_from maf \
  --maf_dir ../MAF_forward \
  --outdir ../SVGAP_run_from_maf \
  --threads 12 \
  --speciesname ISO1_A3_A4_A5 \
  --skip_annotation
```

1.4.3 4.3 Start from aligner outputs

If the inputs are minimap2/LAST/LASTZ/MUMmer/AnchorWave outputs, `01_Convert2Axt.pl` converts them to AXT.

```
perl SVGAP_pipeline.pl \
  --refname ISO1 \
  --glist ../g.list \
  --genome_dir ../genomes \
  --start_from align \
  --aligner minimap2 \
  --align_dir ../alignments \
  --outdir ../SVGAP_run_from_align \
  --threads 12 \
  --speciesname ISO1_A3_A4_A5 \
  --skip_annotation
```

1.4.4 4.4 Use true reciprocal alignments

```
perl SVGAP_pipeline.pl \
  --refname ISO1 \
  --glist ../g.list \
  --genome_dir ../genomes \
  --start_from axt \
  --axt_forward_dir ../AXT_forward \
  --axt_reciprocal_dir ../AXT_query_vs_ref_independent \
  --reciprocal_strategy true \
  --outdir ../SVGAP_run_true_reciprocal \
  --threads 12 \
  --speciesname ISO1_A3_A4_A5 \
  --skip_annotation
```

If no true reciprocal alignments are available, the default `auto` strategy uses forward tnet/qnet and is appropriate.

1.5 5. Main output directories

The pipeline creates the following directories under `--outdir`:

```
SVGAP_run/
  01_axt/                          # Prepared or converted AXT files
  02_chain_forward/                # Forward chain/net/synnet
  02_chain_independent_reciprocal/ # Optional reciprocal chain/net/synnet
  03_filter_forward/               # Filtered synnet + single-coverage MAF
  03_filter_independent_reciprocal/
  04_pairwise_inputs/              # Synnet subsets for PairwiseSV
  04_pairwise_sv/                  # Pairwise raw SVs
  05_combined_sv/                  # Merged DEL/INS/INV/CNV/TRL candidates
  06_genotype/                     # DEL/INS/small-InDel/INV/CXSV genotyping
  07_snp/                          # SNP calling/genotyping
  08_vcf/                          # Final VCF files
  09_annotation/                   # Optional annotation workdir
  SVGAP_pipeline.log               # Main log file
```

Key final VCFs:

```
SVGAP_run/08_vcf/All.largeSV.raw.vcf
SVGAP_run/08_vcf/All.largeSV.annotated.vcf     # if annotation is enabled
SVGAP_run/08_vcf/All.smallIndel.vcf
SVGAP_run/08_vcf/All.sorted.SNP.vcf
SVGAP_run/08_vcf/All.CXSV.vcf
```

1.6 6. Detailed workflow steps and parameters

1.6.1 Step 01: 01_Convert2Axt.pl

Purpose: convert LAST, LASTZ, MUMmer, minimap2, AnchorWave or existing MAF alignments to AXT. It can also split query FASTA into overlapping subcontigs and restore split-contig coordinates to original chromosome coordinates.

```
perl 01_Convert2Axt.pl \
  --ali minimap2 \
  --input ../alignments \
  --wk ../01_axt \
  --tname ISO1 \
  --qname A3 \
  --t 8
```

Option	Meaning
`--ali`	Alignment type: `last`, `lastz`, `maf`, `mummer`, `minimap2`, `anchorwave`.
`--input`	Input alignment directory.
`--wk`	Working/output directory.
`--tname`	Target/reference name. Default: `Ref`.
`--qname`	Query name. Default: `Que`.
`--mode` / `--coord`	`auto`, `chr` or `ctg`. Use `ctg` for split query subcontig alignments.
`--t` / `--threads`	Number of parallel workers. Default: 4.
`--query`	Also generate query-as-reference AXT files.
`--keep_tmp`	Keep temporary files.
`--mum320`	Directory/path for MUMmer `delta2maf`.
`--ucsc`	Directory/path for UCSC utilities such as `mafToAxt`.
`--tba`	Directory/path for TBA utilities.
`--k8`	Path to `k8`.
`--paftools`	Path to `paftools.js`.
`--splitfa` / `--fasta`	FASTA file to split; can be repeated.
`--bin`	Main split window size. Default: 10,000,000 bp.

Option	Meaning
`--slide`	Overlap between windows. Default: 10,000 bp.
`--split_out`	Output directory for split FASTA.
`--sizes` / `--qsize` / `--size`	Original chromosome size file for coordinate restoration in ctg mode.

1.6.2 Step 02: 02_ChainNetSyn.pl

Purpose: build UCSC chain/net/syntenic-net from AXT and optionally generate target-oriented single-coverage MAF.

```
perl 02_ChainNetSyn.pl \
  --gd ../genomes \
  --ad ../01_axt/Target \
  --lst ../g.list \
  --outdir ../SVGAP_run/02_chain_forward \
  --threads 12 \
  --sing
```

Option	Meaning
`--gd`	Genome FASTA directory. The script searches for `ID`, `ID.fa`, `ID.fasta`, or `ID.fna`.
`--ad`	AXT directory. Input names must look like `TargetIDvsQueryID.axt`.
`--lst`	Genome ID list.
`--outdir` / `--syn`	Output directory.
`--ucsc`	UCSC utility directory.
`--tba`	TBA utility directory.
`--linearGap` / `--lg`	ChainNet linearGap file. Default: `pub/TBA/medium`.
`--minscore` / `--minScore`	Minimum score for axtChain/chainNet. Default: 1000.
`--minSpace`	minSpace for chainNet. Default: 1.
`--t` / `--threads`	Number of parallel jobs. Default: 1.
`--sing`	Generate single-coverage MAF.
`--keep_raw_chain`	Keep raw chain files.
`--keep_temp`	Keep temporary AXT/MAF files.
`--force`	Overwrite existing outputs.
`--dry_run`	Print commands without executing them.

1.6.3 Step 03: 03_SynNetFilter.pl

Purpose: filter unreliable short fills in synnet, generate CleanSynNet, raw syntenic-gap files and filtered single-coverage MAF.

```
perl 03_SynNetFilter.pl \
  --synnet ../SVGAP_run/02_chain_forward/synnet \
  --chain ../SVGAP_run/02_chain_forward/chain \
  --gd ../genomes \
  --outdir ../SVGAP_run/03_filter_forward \
  --threads 12
```

Option	Meaning
`--synnet`	Directory containing synnet files from Step 02.
`--chain`	Chain directory from Step 02.

Option	Meaning
`--gd`	Genome FASTA/.sizes/.2bit directory.
`--outdir`	Output directory.
`--ucsc`	UCSC utility directory.
`--tba`	TBA utility directory.
`--cleanSynnet`	Output subdirectory for cleaned synnet. Default: `CleanSynNet`.
`--sing`	Output subdirectory for filtered single MAF. Default: `filtered_sing`.
`--axt`	Output subdirectory for cleaned-synnet AXT.
`--homopairs`	Output subdirectory for homologous pair summaries.
`--rawsv`	Output subdirectory for raw SV gaps.
`--target` / `--refname`	Process only this target/reference.
`--query`	Process only this query.
`--toplen`	Minimum length for top syntenic fills. Default: 500000.
`--nonsynlen`	Minimum length for lower nonSyn fills. Default: 500000.
`--synlen`	Minimum length for lower syn fills. Default: 500000.
`--invlen`	Minimum length for inversion fills. Default: 10000.
`--alrate`	Minimum alignment/fill-length proportion. Default: 0.10.
`--qFarCutoff`	Maximum qFar for inversion fills. Default: 5000000.
`--minscore`	Minimum score threshold.
`--t` / `--threads`	Number of parallel jobs.
`--sing_only_tnet`	Generate single MAF from tnet only.
`--keep_temp`	Keep temporary files.
`--force`	Overwrite existing outputs.
`--dry_run`	Print commands only.

1.6.4 Step 04: 04_PairwiseSV.pl

Purpose: call pairwise SVs from cleaned synnet files, including DEL, INS, CNV, INV, TRL and complex PAV/CMPLX events. This version uses core Perl `fork()/waitpid()` for parallel execution.

```
perl 04_PairwiseSV.pl \
  --syndir ../SVGAP_run/03_filter_forward/CleanSynNet \
  --outdir ../SVGAP_run/04_pairwise_sv \
  --gd ../genomes \
  --threads 12 \
  --seq
```

Option	Meaning
`--synnet` / `--syn`	Single cleaned synnet file.
`--syndir`	Directory of cleaned synnet files.
`--outdir`	Output directory.
`--gd`	Genome FASTA directory.
`--ref_fa`	Reference FASTA for single-file mode.
`--qry_fa` / `--query_fa`	Query FASTA.
`--seq`	Write REF/ALT sequence into output.
`--exact_seq_len`	Try to output full sequences.
`--t` / `--threads`	Number of parallel jobs.
`--invlen`	Minimum inversion length.

Option	Meaning
`--cnvlen`	Minimum CNV length.
`--dellen`	Minimum deletion length.
`--inslen`	Minimum insertion length.
`--trllen`	Minimum translocation length.
`--filltoplen`	Top-fill length threshold.
`--fill2len`	Second-level fill length threshold.
`--rate`	Alignment-rate threshold.
`--max_level`	Maximum synnet parsing level.
`--seq_context`	Flanking context length for sequence output.
`--max_full_seq`	Maximum length for full sequence output.
`--edge_seq_len`	Edge sequence length when long sequences are truncated.
`--max_opposite_len`	Deprecated compatibility option.
`--max_opposite_ratio`	Maximum opposite-side gap ratio for simple DEL/INS classification.
`--keep_tmp`	Keep temporary files.
`--quiet`	Reduce screen output.

1.6.5 Step 05: 05_CombineSV.pl

Purpose: merge pairwise raw SVs into population-level candidate events and split them by SV type.

```
perl 05_CombineSV.pl \
  --sv ../SVGAP_run/04_pairwise_sv \
  --refname ISO1 \
  --outdir ../SVGAP_run/05_combined_sv \
  --sort_threads 12
```

Option	Meaning
`--sv` / `--svdir` / `--pairsv`	Pairwise raw SV directory.
`--refname`	Reference ID.
`--outdir`	Output directory.
`--ins_tol`	Insertion merging tolerance.
`--del_ov`	Minimum deletion overlap for merging.
`--len_ratio`	Allowed length-ratio difference.
`--cant` / `--candidate`	Minimum candidate support.
`--min_large_len`	Minimum large-SV length.
`--max_indel_len`	Maximum small-InDel length.
`--legacy_indel`	Use legacy InDel output logic.
`--write_ins_reference`	Write reference insertion table.
`--cnv_ov`	CNV overlap threshold.
`--cnv_len_ratio`	CNV length-ratio threshold.
`--cnv_bp_tol`	CNV breakpoint tolerance in bp.
`--inv_ov`	Inversion overlap threshold.
`--inv_len_ratio`	Inversion length-ratio threshold.
`--inv_bp_tol`	Inversion breakpoint tolerance in bp.
`--trl_ov`	TRL overlap threshold.
`--trl_len_ratio`	TRL length-ratio threshold.
`--trl_bp_tol`	TRL breakpoint tolerance in bp.
`--trl_require_qchr`	Require query chromosome match for TRL merging.
`--sort_mem`	Memory for sort, for example 2G.
`--sort_threads`	Number of sort threads.
`--sort_bin`	Path to sort binary.

Option	Meaning
`--keep_temp`	Keep temporary files.
`--verbose`	More verbose logging.

1.6.6 Step 06: 06_genotype_DEL.pl

Purpose: genotype merged deletions using single-coverage MAF files.

```
perl 06_genotype_DEL.pl \
  --bed ../SVGAP_run/05_combined_sv/All.DELs.50bplarge.bed.combined.sorted.txt \
  --maf_dir ../SVGAP_run/03_filter_forward/filtered_sing \
  --refname ISO1 \
  --query_list ../SVGAP_run/query.list \
  --out ../SVGAP_run/06_genotype/DEL.large \
  --threads 12
```

Option	Meaning
`--bed`	Merged DEL BED file.
`--maf_dir`	Single-coverage MAF directory.
`--target` / `--refname`	Reference ID.
`--query_list` / `--glist`	Query list; controls VCF sample order.
`--out`	Output prefix.
`--threads` / `--t`	Number of parallel query jobs.
`--maf_suffix`	MAF suffix, usually `.sing.maf`.
`--min_cov`	Minimum reference coverage.
`--min_support`	Minimum alternative support.
`--bridge_del_ratio`	Ratio threshold for bridge-based deletion calls.
`--bridge_ref_low`	Lower bound of bridge reference coverage.
`--bridge_ref_high`	Upper bound of bridge reference coverage.
`--vcf_mode`	VCF output mode.
`--haploid` / `--diploid`	Genotype ploidy.
`--keep_tmp`	Keep temporary per-sample files.

1.6.7 Step 07: 07_genotype_INS.pl

Purpose: genotype merged insertions using MAF context, genome FASTA and sequence/flank matching.

```
perl 07_genotype_INS.pl \
  --bed ../SVGAP_run/05_combined_sv/All.INTs.50bplarge.bed.combined.sorted.txt \
  --maf_dir ../SVGAP_run/03_filter_forward/filtered_sing \
  --genome_dir ../genomes \
  --refname ISO1 \
  --query_list ../SVGAP_run/query.list \
  --out ../SVGAP_run/06_genotype/INS.large \
  --threads 12
```

Option	Meaning
`--bed`	Merged INS BED file.
`--maf_dir`	Single-coverage MAF directory.
`--genome_dir`	FASTA directory for sequence extraction/verification.
`--target` / `--refname`	Reference ID.
`--query_list`	Query list.
`--out`	Output prefix.
`--threads` / `--t`	Number of parallel query jobs.

Option	Meaning
`--maf_suffix`	MAF file suffix.
`--pos_tol`	Position tolerance in bp.
`--seq_id`	Sequence identity threshold.
`--flank`	Flank length for matching.
`--len_ratio_min`	Lower bound of ALT length ratio.
`--len_ratio_max`	Upper bound of ALT length ratio.
`--len_slop`	Absolute length tolerance in bp.
`--min_refcov`	Reference-coverage threshold.
`--block_extra`	Extra length for MAF block search.
`--short_ref_len`	Short reference-anchor threshold.
`--long_seq_len`	Long insertion threshold.
`--sample_len`	Window length for long-sequence sampling.
`--sample_seed_len`	Seed length for sampling.
`--sample_min_hits`	Minimum seed hits.
`--sample_windows`	Number of sampling windows.
`--vcf_mode`	VCF output mode.
`--haploid` / `--diploid`	Genotype ploidy.
`--bed_one_based`	Treat input BED as 1-based.
`--alt_is_insseq`	Treat ALT as the inserted sequence itself.
`--keep_tmp`	Keep temporary files.

1.6.8 Step 08: 08_genotype_small_indels.pl

Purpose: genotype 1-50 bp small insertions and deletions. It can read small DEL and small INS files together.

```
perl 08_genotype_small_indels.pl \
  --maf_dir ../SVGAP_run/03_filter_forward/filtered_sing \
  --refname ISO1 \
  --query_list ../SVGAP_run/query.list \
  --del_bed ../SVGAP_run/05_combined_sv/All.DELs.50bpsmall.bed.combined.sorted.txt \
  --ins_bed ../SVGAP_run/05_combined_sv/All.INTs.50bpsmall.bed.combined.sorted.txt \
  --out ../SVGAP_run/06_genotype/smallIndel \
  --threads 12
```

Option	Meaning
`--maf_dir`	Single-coverage MAF directory.
`--target` / `--refname`	Reference ID.
`--bed` / `--variants`	Mixed small-InDel BED file.
`--del_bed`	Small deletion BED file.
`--ins_bed`	Small insertion BED file.
`--out`	Output prefix. Default: `small_indel_genotype`.
`--max_len`	Maximum InDel length. Default: 50.
`--min_len`	Minimum InDel length. Default: 1.
`--maf_suffix`	MAF suffix. Default: `.sing.maf`.
`--query_list` / `--glist`	Query list.
`--other_gt`	GT for non-reference/non-alternative alleles. Default: `./.`.
`--use_block_gap` / `--no-use_block_gap`	Infer genotypes from adjacent MAF block gaps. Default: on.
`--max_junction_gap`	Maximum junction-gap length. Default: 1000.
`--threads` / `--p`	Number of parallel query jobs.
`--keep_tmp`	Keep temporary files.
`--bed_coord`	Coordinate system: 0 for 0-based MAF/BED, 1 for 1-based.

Option	Meaning
`--gt_pos_slop`	Position tolerance for relaxed genotyping. Default: 5.
`--len_slop`	Length tolerance for relaxed genotyping. Default: 0.
`--relaxed_gt` / `--no-relaxed_gt`	Search nearby MAF gaps before calling reference. Default: on.

1.6.9 Step 09: 09_SNPgenotyping.pl

Purpose: call SNPs from single-coverage MAF and merge all query SNP sites into one population VCF.

```
perl 09_SNPgenotyping.pl \
  --singmaf ../SVGAP_run/03_filter_forward/filtered_sing \
  --refsize ../genomes/ISO1.sizes \
  --refname ISO1 \
  --query_list ../SVGAP_run/query.list \
  --speciesname ISO1_A3_A4_A5 \
  --outdir ../SVGAP_run/07_snp \
  --threads 12
```

Option	Meaning
`--singmaf`	Single-coverage MAF directory or file.
`--refsize`	Reference chromosome size file.
`--refseq`	Kept for compatibility; usually not required.
`--refname` / `--target`	Reference ID.
`--query_list` / `--glist`	Query list controlling VCF sample order.
`--speciesname`	Project/species name for VCF header.
`--outdir`	Output directory.
`--t` / `--threads`	Number of parallel jobs.
`--tmpdir`	Temporary directory.
`--keep_tmp`	Keep temporary files.
`--allow_ambiguous`	Allow ambiguous bases.
`--minNS`	N/ambiguous filter threshold.
`--phased`	Output phased genotypes.
`--legacy_format`	Use legacy format.
`--force`	Overwrite existing outputs.

1.6.10 Step 10: 10_SVannotation.pl

Purpose: annotate SV VCF records, including tandem duplication, TE overlap and CDS overlap. For symbolic DEL alleles, the deleted sequence can be extracted from reference coordinates.

```
perl 10_SVannotation.pl \
  --ref ../genomes/ISO1.fa \
  --vcf ../SVGAP_run/08_vcf/All.largeSV.raw.vcf \
  --te ../annotation/TE.bed \
  --cds ../annotation/CDS.bed \
  --out ../SVGAP_run/08_vcf/All.largeSV.annotated.vcf \
  --workdir ../SVGAP_run/09_annotation \
  --threads 12 \
  --minimap2 minimap2
```

If minimap2 is unavailable, skip annotation in the main pipeline:

```
--skip_annotation
```

or skip tandem-duplication detection when running annotation alone:

`--skip_tandem_dup`

Option	Meaning
`--ref`	Reference FASTA.
`--vcf`	Input VCF.
`--te`	TE annotation file.
`--cds`	CDS annotation file.
`--out`	Output VCF.
`--svtype`	Annotate only a selected SV type.
`--workdir`	Working directory.
`--threads` / `--t`	Number of threads.
`--minimap2`	Path to minimap2.
`--kmer`	K-mer size for tandem-duplication seeds.
`--dup_seed_fraction`	Seed hit fraction for tandem duplication.
`--skip_tandem_dup`	Skip tandem-duplication detection.
`--min_annot_pct`	Minimum TE/CDS overlap percentage.
`--min_aln_len`	Minimum minimap2 alignment length.
`--min_mapq`	Minimum minimap2 MAPQ.
`--max_dup_flank`	Maximum flank length for tandem-duplication checking.
`--max_dup_seeds`	Maximum number of duplication seeds.
`--max_seq_len`	Maximum sequence length used for annotation.
`--te_preset`	TE input preset.
`--cds_preset`	CDS input preset.
`--keep_tmp`	Keep temporary files.
`--quiet`	Reduce output.

1.6.11 Step 11: 11_INVgenotyping.pl

Purpose: genotype merged inversion candidates using MAF body coverage and flank coverage.

```
perl 11_INVgenotyping.pl \
  --inv ../SVGAP_run/05_combined_sv/All.INVs.bed.combined.sorted.txt \
  --maf_dir ../SVGAP_run/03_filter_forward/filtered_sing \
  --refsize ../genomes/ISO1.sizes \
  --refname ISO1 \
  --speciesname ISO1_A3_A4_A5 \
  --query_list ../SVGAP_run/query.list \
  --out ../SVGAP_run/06_genotype/INV.vcf \
  --threads 12
```

Option	Meaning
`--inv`	Inversion BED file.
`--maf_dir`	Single-coverage MAF directory.
`--sing` / `--singmaf`	Single MAF input or directory.
`--refsize`	Reference size file.
`--refname` / `--target`	Reference ID.
`--speciesname`	Project name for VCF header.
`--query_list` / `--glist`	Query list.
`--out`	Output VCF.
`--outdir`	Output directory.
`--maf_suffix`	MAF suffix.
`--threads` / `--t`	Number of parallel query jobs.
`--bed_coord`	BED coordinate system.
`--flank_ratio`	Flank length as a fraction of inversion length.

Option	Meaning
`--min_flank`	Minimum flank length.
`--max_flank`	Maximum flank length.
`--min_body_cov`	Minimum body coverage.
`--min_flank_cov`	Minimum flank coverage.
`--dominant_frac`	Minimum dominant-orientation fraction.
`--min_body_bp`	Minimum body coverage in bp.
`--trust_carriers`	Trust candidate carrier samples.
`--body_only`	Use body evidence only.
`--phased`	Output phased genotypes.
`--keep_tmp`	Keep temporary files.
`--debug`	Output debug information.

1.6.12 Step 12: 12_combine_genotype_CXSV.pl

Purpose: merge and genotype complex SVs where reference and query gaps coexist and cannot be represented well as simple DEL/INS.

```
perl 12_combine_genotype_CXSV.pl \
  --refname ISO1 \
  --query_list ../SVGAP_run/query.list \
  --rawsv_dir ../SVGAP_run/04_pairwise_sv \
  --outdir ../SVGAP_run/06_genotype/CXSV \
  --break_tolerance 2000 \
  --min_ro 0.50 \
  --len_ratio 1.00
```

Option	Meaning
`--refname`	Reference ID.
`--glist`	Genome list.
`--query_list`	Query list; preferred for sample order.
`--rawsv_dir` / `--svdir`	Step 04 raw SV directory.
`--outdir`	Output directory.
`--break_tolerance`	Breakpoint merging tolerance. Default: 2000.
`--min_ro`	Minimum reciprocal overlap. Default: 0.50.
`--len_ratio`	Maximum relative length difference. Default: 1.00.
`--min_component_size`	Minimum number of component fragments.
`--unknown_missing`	Output `./.` instead of `0|0` for non-carriers.
`--keep_ref_sample`	Keep reference sample in VCF.
`--max_info_source_ids`	Maximum number of source IDs kept in INFO.

1.6.13 SVGAP_pipeline.pl controller options

Option	Meaning
`--refname` / `--target`	Reference genome ID; must appear in `g.list`.
`--glist`	Genome list; VCF sample order is the list after removing the reference.
`--genome_dir` / `--gd`	Genome FASTA directory.
`--start_from`	Start point: `align`, `maf`, or `axt`. Default: `align`.
`--align_dir` / `--input`	Aligner output directory.
`--aligner` / `--ali`	Aligner type: `minimap2`, `last`, `lastz`, `mummer`, `anchorwave`, `maf`.
`--maf_dir`	Forward MAF directory.

Option	Meaning
`--maf_reciprocal_dir`	True independent reciprocal MAF directory.
`--axt_dir` / `--ad`	Forward AXT directory.
`--axt_forward_dir`	Forward AXT directory.
`--axt_reciprocal_dir`	True independent reciprocal AXT directory.
`--reciprocal_strategy`	`auto`, `qnet`, `true`, or `both`. Default: `auto`.
`--outdir`	Main output directory. Default: `SVGAP_run`.
`--threads` / `--t`	Number of threads/jobs.
`--speciesname`	Project/species name in VCF headers.
`--ucsc`	UCSC utility path.
`--tba`	TBA utility path.
`--k8`	k8 path.
`--paftools`	paftools.js path.
`--mum320`	MUMmer/delta2maf path.
`--linearGap` / `--lg`	ChainNet linearGap file.
`--refsize`	Reference size file.
`--te`	TE annotation file.
`--cds`	CDS annotation file.
`--minimap2`	minimap2 path.
`--annotation` / `--no-annotation`	Enable/disable annotation.
`--skip_convert`	Skip Step 01.
`--skip_chainnet`	Skip Step 02.
`--skip_filter`	Skip Step 03.
`--skip_pairwise_sv`	Skip Step 04.
`--skip_combine`	Skip Step 05.
`--skip_genotype`	Skip DEL/INS/small-InDel/INV genotyping.
`--skip_cxsv`	Skip Step 12.
`--skip_snp`	Skip Step 09.
`--skip_annotation`	Skip Step 10.
`--cxsv_break_tolerance`	CXSV breakpoint merging tolerance.
`--cxsv_min_ro`	CXSV minimum reciprocal overlap.
`--cxsv_len_ratio`	CXSV length-ratio threshold.
`--cxsv_unknown_missing`	Output missing genotypes for CXSV non-carriers.
`--dry_run`	Print commands without running.
`--force`	Overwrite existing outputs.

1.6.14 SVGAP_vcf_tools.pl

Purpose: merge multiple VCFs and enforce sample order according to `query.list` or `g.list`.

```
perl SVGAP_vcf_tools.pl \
  --mode merge \
  --vcfs DEL.vcf,INS.vcf,INV.vcf \
  --query_list query.list \
  --refname ISO1 \
  --refsize ../genomes/ISO1.sizes \
  --out All.largeSV.raw.vcf
```

Option	Meaning
`--mode`	Current main mode is `merge`.
`--vcfs`	Comma-separated VCF paths.
`--out`	Output VCF.
`--query_list` / `--glist`	File controlling sample order.
`--refname` / `--target`	Reference ID.
`--refsize`	Chromosome size file for sorting.

Option	Meaning
`--missing_gt`	Missing genotype. Default: `./.`

1.7 7. Output sanity checks

After the run, inspect:

```
ls -lh SVGAP_run/08_vcf/
less SVGAP_run/SVGAP_pipeline.log
```

Check VCF sample columns:

```
grep '^#CHROM' SVGAP_run/08_vcf/All.largeSV.raw.vcf
```

Expected order:

```
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT A3 A4 A5
```

Count records:

```
grep -v '^#' SVGAP_run/08_vcf/All.largeSV.raw.vcf | wc -l
grep -v '^#' SVGAP_run/08_vcf/All.smallIndel.vcf | wc -l
grep -v '^#' SVGAP_run/08_vcf/All.sorted.SNP.vcf | wc -l
grep -v '^#' SVGAP_run/08_vcf/All.CXSV.vcf | wc -l
```

1.8 8. Troubleshooting

1.8.1 8.1 Parallel::ForkManager warning still appears

First confirm you are running the clean package, not an old copy:

```
pwd
head -5 SVGAP_pipeline.pl
```

The script should contain:

```
# Author: Yi Liao @ scau.edu.cn
```

and:

```
use lib "$Bin/lib";
```

Then test:

```
perl -Ilib -MParallel::ForkManager -e 'print "OK\n"'
```

If this works but the pipeline still warns, the most likely cause is that the command is calling an older script path.

1.8.2 8.2 AXT files are not detected

AXT files must be non-empty and named:

```
ISO1vsA3.axt
ISO1vsA4.axt
ISO1vsA5.axt
```

Zero-byte files are ignored.

1.8.3 8.3 Genome FASTA not found

The ID in `g.list` must match the FASTA prefix. For `A3`, one of the following must exist:

```
A3
A3.fa
A3.fasta
A3.fna
```

1.8.4 8.4 minimap2 is missing

The core SV/SNP/InDel workflow does not always require system minimap2, but annotation can require it for tandem-duplication checks. If minimap2 is unavailable, use:

`--skip_annotation`

or, for annotation alone:

`--skip_tandem_dup`

1.8.5 8.5 Memory and threads

Higher `--threads` can speed up query-level parallelism but also increases memory use. For three queries, `--threads 3` to `--threads 12` is usually reasonable. For large genomes, start with `--threads 4` or `--threads 6`.

1.8.6 8.6 qnet versus reciprocal alignments

If no true query-vs-reference alignment exists, keep the default `auto` behavior. SVGAP will use forward chainNet tnet/qnet to recover reference insertion information. If true independent `A3vsIS01.axt`, `A4vsIS01.axt` and `A5vsIS01.axt` files exist, use `--reciprocal_strategy true`.

1.9 9. Recommended training workflow

For a training session:

1. Show `g.list` and genome FASTA naming.
2. Show AXT naming: `IS01vsA3.axt`.
3. Run the controller with `--dry_run` so trainees can see each command.
4. Run the full pipeline first with `--skip_annotation`.
5. Focus on the four outputs in `08_vcf/`: large SV, small InDel, SNP and CXSV.
6. Explain how `query.list` controls VCF sample order.
7. Demonstrate annotation and benchmark tools separately.

Dry-run example:

```
perl SVGAP_pipeline.pl \
  --refname IS01 \
  --glist ../g.list \
  --genome_dir ../genomes \
  --start_from axt \
  --axt_dir ../01_axt/Target \
  --outdir ../SVGAP_run_dryrun \
  --threads 12 \
  --skip_annotation \
  --dry_run
```